

**Proposed Data Centre
Development at No. 2-10,
Tai Yuen Street, Kwai
Chung, N.T.**

Technical Particular
Specifications for Excavation

2020-10-19

Contents

1.1	Nature of excavation.....	1
1.2	Site Levels	1
1.3	Generals	1
1.4	Obstructions met during excavations	1
1.5	Planking and strutting	1
1.6	Bottoms of Excavations	2
1.7	Damage to existing drains, water mains etc.	2
1.8	Excavation below required levels	2
1.9	Return fill in and ram	2
1.10	Surplus excavated material	2
1.11	Site core-stone, boulders, rock, etc.	3
1.12	Site, excavations etc. free from water	3
1.13	Storm water drainage	3
1.14	Hardcore	3
1.15	Building paper.....	3
1.16	Chunam surfacing	3
2.1	Filling material and method	4
2.2	Acceptance Criteria	4
2.3	Tolerance for moisture content.....	4
2.4	Test for determining the degree of compaction of compacted fill	4
2.5	Filling of foundation pits and trenches.....	5

1. Excavation

1.1 Nature of excavation

- 1.1.1 The Contractor shall be deemed to have inspected the Site and understand the existing site conditions such as site levels and adjoining structures, prior to tendering, to form his own opinion of the nature of the ground to be excavated and to estimate the percentage of rock, boulders, old foundations, etc to be removed. Such opinion and estimate shall be at the Contractor's sole risk and will not be subjected to any adjustment.
- 1.1.2 Information which the Architect may possess about the nature of the soil from trial boreholes etc. will be made available on request.
- 1.1.3 This information will be given in good faith but no guarantee can be given of its accuracy or applicability over the whole site.
- 1.1.4 Any additional preliminary investigation may be carried out by the Contractor at his own expense.

1.2 Site Levels

- 1.2.1 All site levels either spot or contour shown on the drawings are, unless otherwise stated, reduced to Principal Datum.
- 1.2.2 If the Contractor is not satisfied with the accuracy of these levels, he shall give written notice to the Architect before excavations are commenced, otherwise no claim in respect of the accuracy of such levels will be considered.

1.3 Generals

- 1.3.1 The Main Contractor shall excavate for foundation, drains, water pipes, etc. and to reduce levels, etc. to such extent and levels as shown on the Drawings or as directed by the Architect, provide sufficient hard dry material for refilling where necessary and well rammed before concreting.

1.4 Obstructions met during excavations

- 1.4.1 All roots, buried tree trunks, old steel plate, existing precast pile, old concrete, reinforced concrete or brick foundations, non-serviceable drains, manholes and gullies, or other obstructions met with during the course of the work shall be removed. The open ends of all such drains, etc. shall be sealed with weak concrete and all voids filled in with hard dry material well rammed and consolidated.

1.5 Planking and strutting

- 1.5.1 The Contractor shall be responsible for the design, supply, erection and removal of all planking and strutting or shoring that may be necessary for the proper execution of the excavation works.
- 1.5.2 The Contractor shall be responsible for the security and stability of all such planking and strutting or shoring.
- 1.5.3 The design of all such planking and strutting or shoring shall be submitted to the Architect and Building Authorities for approval, but the approval or comments given by the Architect or Building Authorities of the Contractor's design or the adaptation by the Contractor on the Architect's design on planking and strutting or shoring shall not relieve in any way the Contractor's responsibility as aforesaid.
- 1.5.4 Unless otherwise stated, all planking and strutting or shoring shall be removed on completion of the foundation works.

1.6 Bottoms of Excavations

- 1.6.1 The bottoms of the excavations shall be formed to correct levels and falls as required and well rammed and consolidated.
- 1.6.2 All water and sludge shall be removed, and the bottoms of excavations shall be perfectly dry before any concrete, hardcore, drain pipe, etc. is laid.
- 1.6.3 All excavations shall be inspected and approved by the Architect before any concrete, hardcore, drain pipe, etc. is laid. If the levels or depths are not consistent the drawings, permission in writing shall be obtained from the Architect before proceeding further with the excavation.

1.7 Damage to existing drains, water mains etc.

- 1.7.1 Care shall be taken not to disturb, or damage existing and serviceable drains, gas or water mains, cables or other underground work met with during the course of the work. The Contractor shall either provide adequate temporary supports to same during excavation or temporarily divert same and reinstate afterwards. Should any damage occur, the Architect and the Authority concerned shall be notified immediately and all damage made good at the Contractor's expense.

1.8 Excavation below required levels

- 1.8.1 Should the Contractor excavate below the levels shown on the drawings without the Architect's written instructions, he shall, at his own expense, fill in the excess excavations with C45/20 concrete or as instructed by the Architect.
- 1.8.2 Should the Contractor allow water and sludge to accumulate to such an extent that it has to be excavated out to a dry and firm bottom, the Contractor shall carry out such further excavation as may be ordered by the Architect and fill this excess excavation with such concrete as instructed by the Architect all at his own expense.

1.9 Return fill in and ram

- 1.9.1 The Contractor shall return and fill in selected excavated material around foundations up to original ground level or to make up levels underground slabs, paving, etc. in layers not exceeding 225mm thick, well watered and carefully rammed and consolidated.
- 1.9.2 No filling shall be executed until the foundations have been inspected and approved by the Architect or his representative.
- 1.9.3 Filling material shall be free from top-soil, slurry, perishable materials, organic matter and other deleterious substances.
- 1.9.4 Unless otherwise required by the Contract, the natural ground over which filling is to be placed shall be cleared of all loose boulders, grass, productive soil, bushes, trees, roots and other vegetation. The surface of sloping ground shall be benched in steps as specified which shall be trimmed and compacted to form relatively firm and level surfaces to receive the fill.

1.10 Surplus excavated material

- 1.10.1 All surplus excavated material shall be removed off site, the Contractor being responsible for finding a dump and for paying all fees and charges.

1.11 Site core-stone, boulders, rock, etc.

- 1.11.1 All core-stone boulders, rock that is excavated from the site shall be disposed of by the Contractor. They may if approved by the Architect be broken for use as aggregate, hardcore or masonry, otherwise they shall be removed at the Contractor's own expense.

1.12 Site, excavations etc. free from water

- 1.12.1 The Contractor shall keep the whole of the site, excavations etc. from water from whatever source at all times by pumping, bailing or other means at his own expense. No accumulation of water will be permitted at any time during the Contract Period.
- 1.12.2 All areas where concrete is to be placed must be reasonably dry at the time of concrete pouring and until such time as the concrete work has gained sufficient strength to resist the forces arising from the water.

1.13 Storm water drainage

- 1.13.1 The Contractor shall ensure that storm water does not flow on to public roads or adjoining properties and shall make temporary provision for disposal of storm water during the carrying out of the Works by temporary surface channels and other suitable means which shall either be removed from, all work disturbed being made good, or be left on the site on completion of the Works, as directed by the Architect.

1.14 Hardcore

- 1.14.1 Hardcore shall be composed of clean, dry, broken brick, stone or concrete, broken to 50mm to 75mm gauge, spread and levelled and well rammed and consolidated to the thicknesses required.
- 1.14.2 The upper surface shall be blinded with a 25mm layer of sand or red earth watered, rolled and tamped into the interstices of the hardcore to provide a fair and partly impervious surface.
- 1.14.3 Hardcore beds under paving subject to vehicular traffic shall be rolled with a ten tonne roller until fully compacted and a dense close-knit surface is obtained; fine material shall be added if necessary.

1.15 Building paper

- 1.15.1 The Contractor shall lay under concrete surface beds, a layer of "Sisalcraft" or other approved sub-soil grade reinforced waterproof building paper or, alternatively, 0.08mm thick polyethylene sheeting, lapped 150mm and taped at all joints to prevent leakage of the water content of the concrete and sealed around pipes etc. penetrating the membrane.

1.16 Chunam surfacing

- 1.16.1 Chunam for surfacing shall be composed of the following materials in the proportions by weight:
- (1) 1 part Portland cement,
 - (2) 3 parts hydrated lime, and
 - (3) 20 parts approved non-organic soil
- 1.16.2 The materials shall be well-mixed, well compacted, sprinkled with water and finished to a uniform final surface.
- 1.16.3 Before chunam is applied, the exposed soil slope shall be cleared of all debris, loose material and vegetation.

- 1.16.4 Joints between successive days' work shall be neat and straight, and panels shall be rectangular where possible.
- 1.16.5 Just before the chunam hardens properly, it shall be thoroughly wetted and rubbed with sacks to fill any cracks which have appeared.
- 1.16.6 The finished surface shall be kept wet for such curing period as instructed by the Architect.
- 1.16.7 50mm diameter weepholes shall be built in slopes at 1.50m centres staggered in each direction as the work proceeds. The minimum length of weepholes shall be 100mm.

2. Backfill

2.1 Filling material and method

- 2.1.1 The backfilling material and backfilling method shall be proposed by contractor and fully comply with General Specification General Specification for Building 2017 Edition and its corrigendum published by Architectural Services Department. Should the following particular specifications have any deviations from the general specification, the contractor shall refer to general specification.
- 2.1.2 Unless otherwise specified, backfilling consists of general filling material obtained from excavation on Site, borrow areas and other approved sources.
- 2.1.3 Provide a method statement showing the sources of fill for each fill area, the constructional plant to be used for placing and the method of compaction.
- 2.1.4 Obtain approval before commencing filling and before any fill layer is covered.
- 2.1.5 Agree with the Architect the thickness of each layer which shall be compatible with the compaction plant used. Carry out carefully controlled tests to determine the optimum placing thickness for the particular fill and the number of passes to achieve the required density with the compaction plant provided. No permanent fill shall be placed until such tests have been agreed with the Architect. Earth moving plant will not be accepted as compaction equipment under this clause. Spread material in layers of uniform thickness and compact as soon as practicable after deposition.

2.2 Acceptance Criteria

- 2.2.1 Unless otherwise stated in the Contract, fill material shall be compacted to obtain a relative compaction of at least 95% throughout. Relative compaction is determined in accordance with Clause 2.4 below.

2.3 Tolerance for moisture content

- 2.3.1 The moisture content of the in-situ material during compaction shall be within $\pm 3\%$ of the optimum moisture content determined in accordance with Clause 2.4 below. Add water to soil in the stockpile if required before spreading in layers.

2.4 Test for determining the degree of compaction of compacted fill

- 2.4.1 Arrange for tests specified below to be carried out by an approved independent testing firm, using their own equipment and provide such labour and assistance as may be required.
- 2.4.2 Determine the maximum dry density and optimum moisture content in accordance with GEO Report No. 36. Test each soil type when first used and thereafter at the same time as every set of field density tests is carried out.

- 2.4.3 Determine the in-situ field density and moisture content in accordance with GEO Report No.36 (Sand Replacement) or with ASTM D-2167 63T (Densimeter) to determine the relative compaction achieved. Test each layer and obtain approval before the next layer is placed. Unless otherwise stated in the Contract, the number of tests required shall follow the General Specification for Civil Engineering Works - Table 6.5. Keep records identifying the soil type and location in the Works showing the following for each series of tests:
- (1) Dry density of soil tested
 - (2) Moisture content
 - (3) Relative compaction achieved (%)
- 2.4.4 Determine moisture content in accordance with GEO Report No. 36 by means of a drying-oven. In order to expedite results, moisture contents for field density tests may be determined by use of a rapid moisture tester using carbide. However, this should not be used until it has been calibrated against moisture content tests carried out in a drying oven. Submit a calibration chart for approval before a rapid moisture tester is used.
- 2.4.5 Determine moisture content in accordance with GEO Report No. 36 by means of a drying-oven. In order to expedite results, moisture contents for field density tests may be determined by use of a rapid moisture tester using carbide. However, this should not be used until it has been calibrated against moisture content tests carried out in a drying oven. Submit a calibration chart for approval before a rapid moisture tester is used.

2.5 Filling of foundation pits and trenches

- 2.5.1 Deposit and compact suitable filling material to achieve the compaction specified above unless otherwise shown on the drawings or directed by the Architect. Carefully remove planking and strutting and other excavation supports as the filling proceeds except where they are specified or required by the Architect to be left in position.

Document prepared by

Aurecon Hong Kong Limited

Unit 1608, 16/F, Manulife Financial Centre

223-231 Wai Yip Street

Kwun Tong

Kowloon, Hong Kong

T +852 3664 6888

F +852 3664 6999

E hongkong@aurecongroup.com

W aurecongroup.com

aurecon

*Bringing ideas
to life*

Aurecon offices are located in:

Angola, Australia, Botswana, China,
Ghana, Hong Kong, Indonesia, Kenya,
Lesotho, Macau, Mozambique,
Namibia, New Zealand, Nigeria,
Philippines, Qatar, Singapore, South Africa,
Swaziland, Tanzania, Thailand, Uganda,
United Arab Emirates, Vietnam.